

VISION Resistivity
Dual Frequency
Recorded Mode

Schlumberger

Country: Norway

Company: Equinor

Well: 31/5-7

Field: Eos

Rig Name: West Hercules

Country: Norway

Latitude: 60° 34' 35.14" N

UWID:

Longitude: 3° 26' 36.13" E

Rig Name:

West Hercules

Block: 31/5

Rig Type:

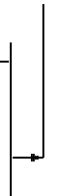
Semi-Submersible

FL: Norwegian North Sea

FL1: Section: 12 1/4 x 17 1/5 in.

FL2: Job no.: 19SCA0085

Log Measured From: - Drill Floor: 31.00 m
Permanent Datum: - Mean Sea Level



Ground Level: 307.00 m

Acquisition Dates: 19-Dec-2019 -- 20-Dec-2019

Other Services:

Log Interval: 919.00(m) -- 1908.00(m)

Direction & Inclination (Surveys)

Index Types: Measured Depth

Annular Pressure While Drilling

Index Scales: 1:500

Depth Source: Driller's Depth

Depth Sensor: 3rd Party Depth

Print Type: Final

Spud Date: 02-Dec-2019



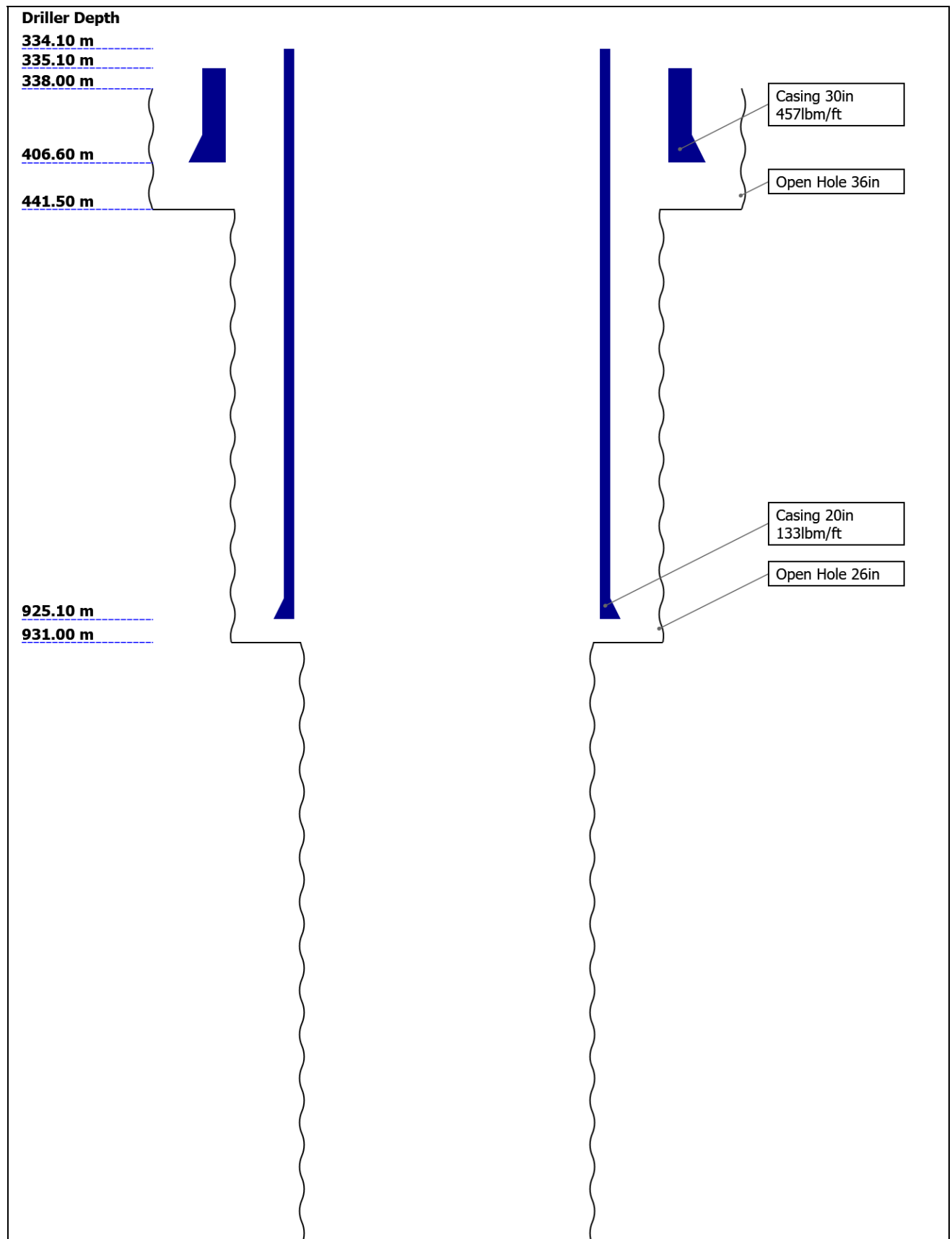
Disclaimer

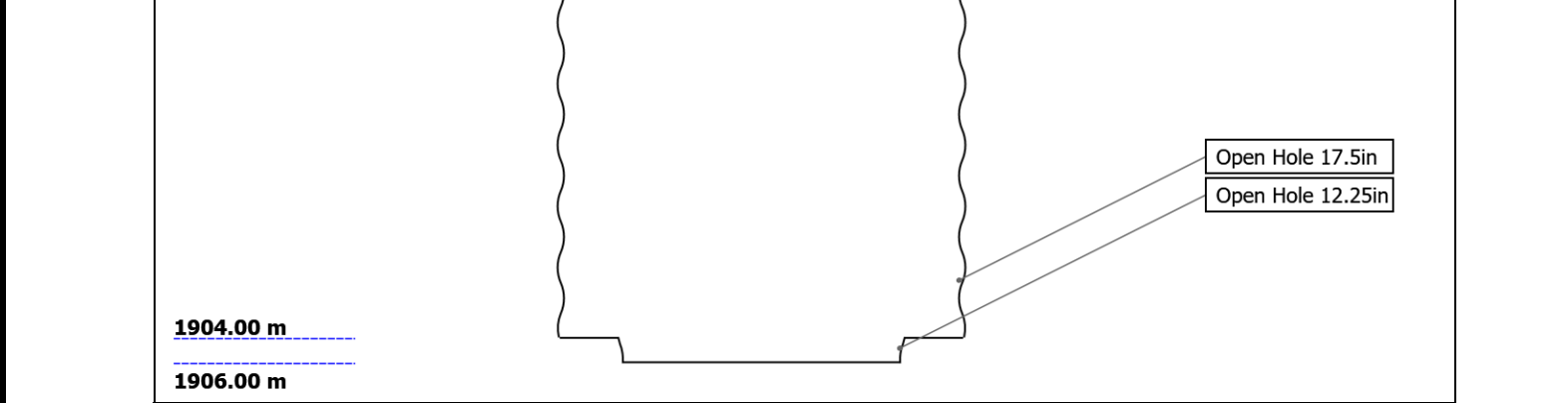
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Well Sketch





Borehole Size/Casing Record						
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Bit						
Bit Size (in)	36	26	17.5	12.25		
Top Driller (m)	338	441.5	931	1904		
Bottom Driller (m)	441.5	931	1904	1906		
Casing						
Size (in)	30	20				
Weight (lbm/ft)	457	133				
Inner Diameter (in)	27.067	18.73				
Grade	X52	N80				
Top Driller (m)	335.1	334.1				
Bottom Driller (m)	406.6	925.1				

Operational Run Summary						
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Parameter (unit)	Run 7					
Date Log Started	19-Dec-2019					
Time Log Started	04:52:35					
Date Log Finished	20-Dec-2019					
Time Log Finished	22:18:15					
Bit Size (in)	17.500					
Bit Start Depth (m)	931.00					
Bit Stop Depth (m)	1906.00					
Top Log Interval (m)	931.00					
Bottom Log Interval (m)	1894.18					
Max Hole Deviation (deg)	1.10					
Azimuth of Max Deviation (deg)	133.27					
Logging Unit Number	A615					
Logging Unit Location	Service Office					
Recorded By	G.Boyd					
Witnessed By	S.Vikra					
Service Order Number	19SCA0085					

Borehole Fluids						
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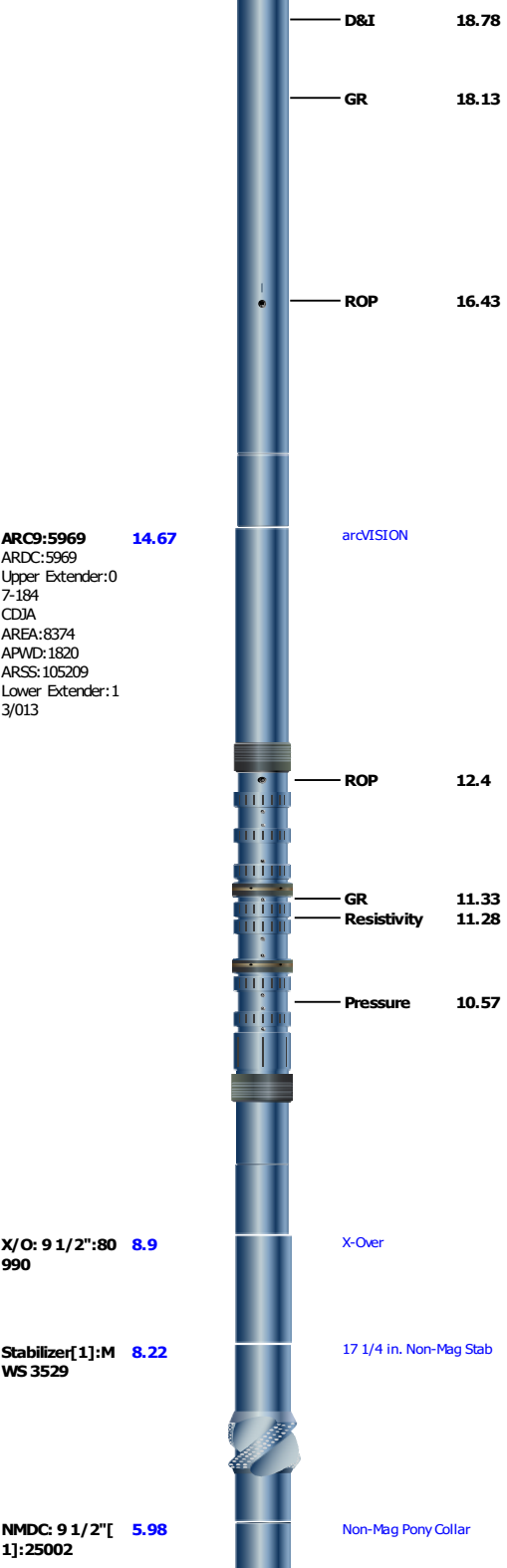
Parameter(unit)	Run 7					
Fluid Type	Water					
Fluid Name	Glydril					
Max Recorded Temperatures (degC)	38					
Source of Sample	Active Tank					
Salinity (ppm)	96819.84					
Density (g/cm3)	Zoned					
Funnel Viscosity (s)						
Fluid Loss (cm3)						
PH						
Source RMF						
RMC	Pressed					
RM @ Meas Temp (ohm.m@degC)	0.09 @ 16.3					
RMF @ Meas Temp (ohm.m@degC)	0.06 @ 16					
RMC @ Meas Temp (ohm.m@degC)	0.12 @ 20					
RM @ BHT (ohm.m@degC)	0.06 @ 38					
RMF @ BHT (ohm.m@degC)	0.04 @ 38					
RMC @ BHT (ohm.m@degC)	0.08 @ 38					
Total Solid (%)						
High Gravity Solids (%)						

Run 7

Parameter	Value	Start
Density	1.25	12/19/2019 4:52:35 AM
Density	1.26	12/19/2019 5:00:00 PM

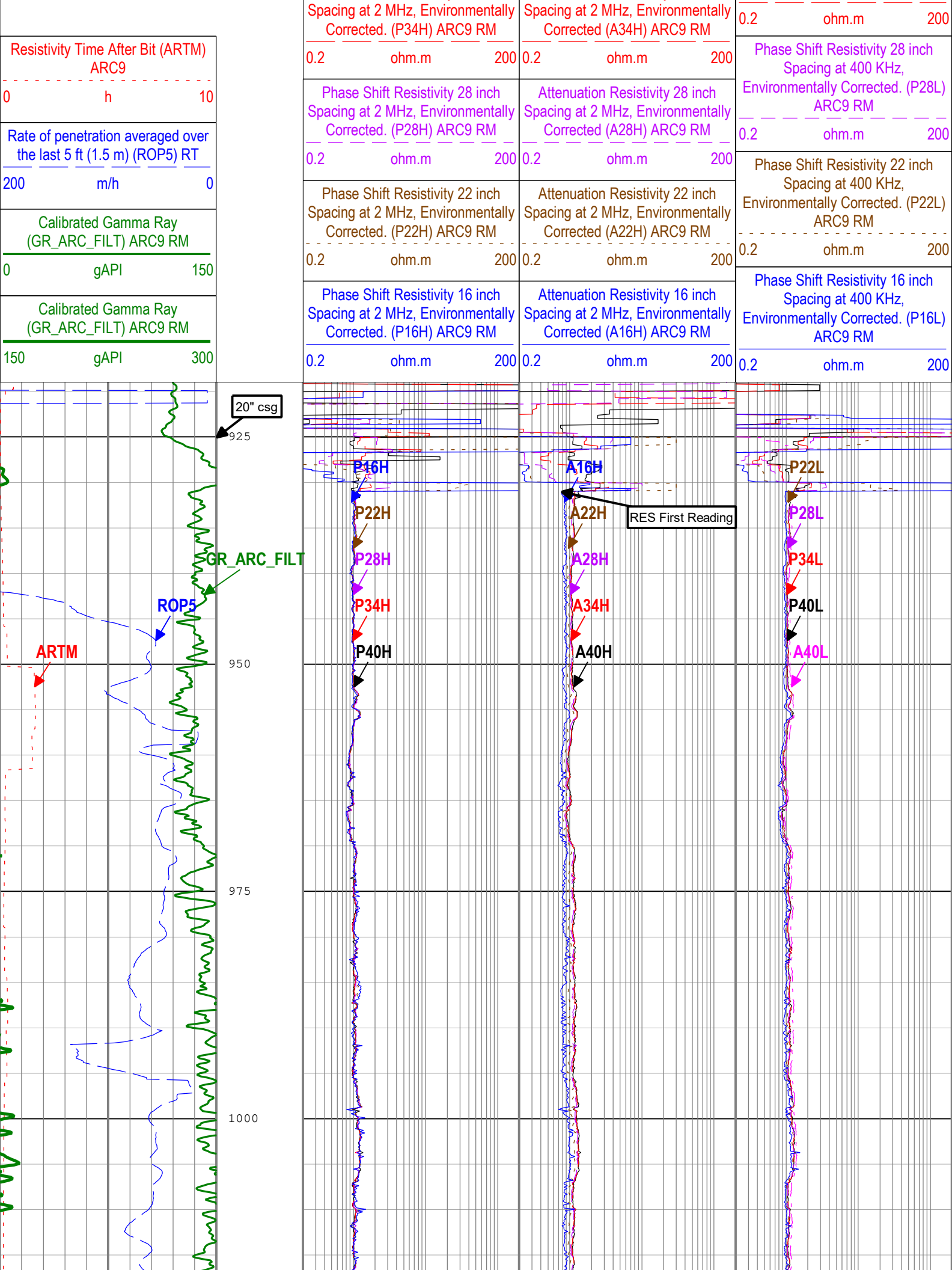
Remarks and Equipment Summary

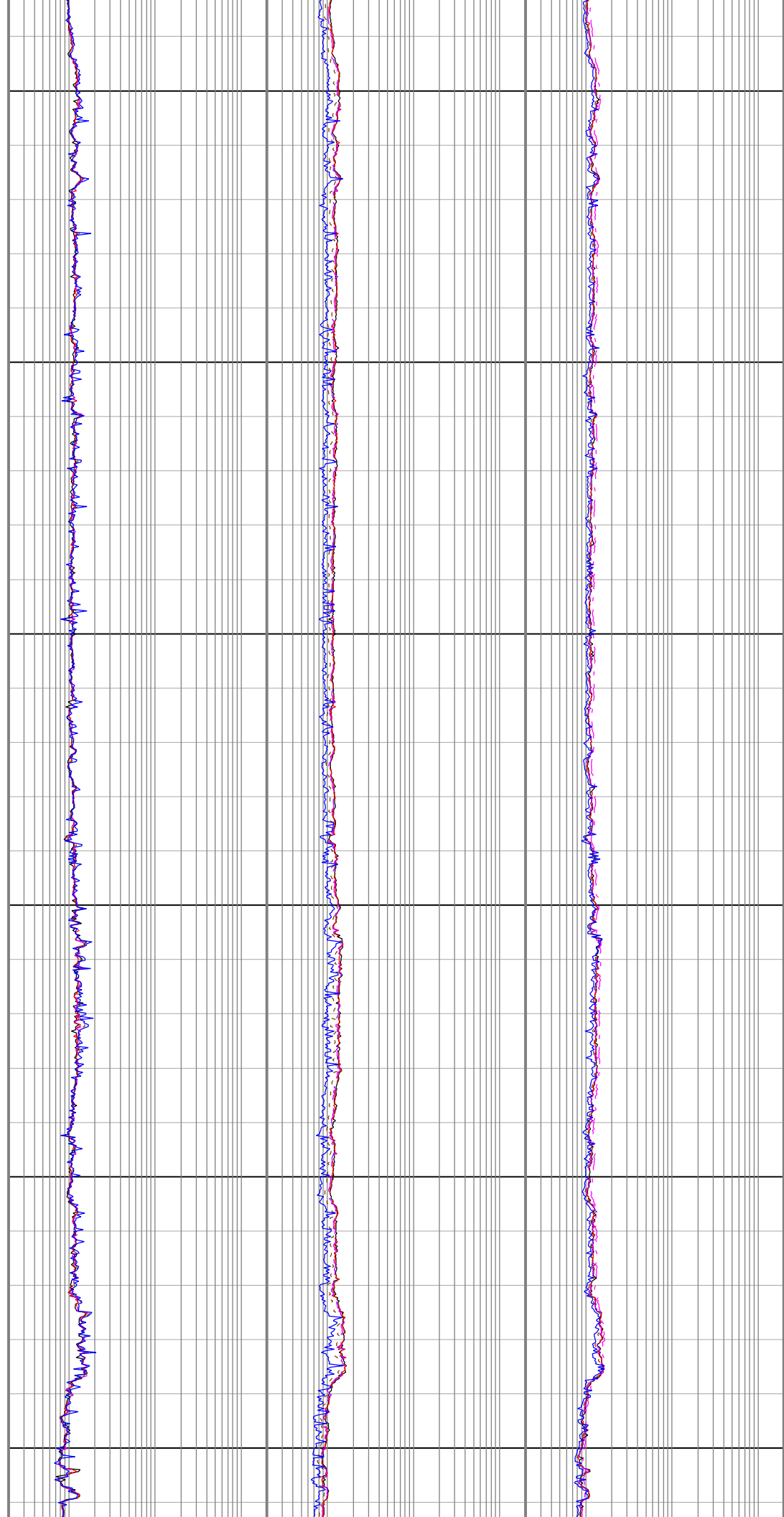
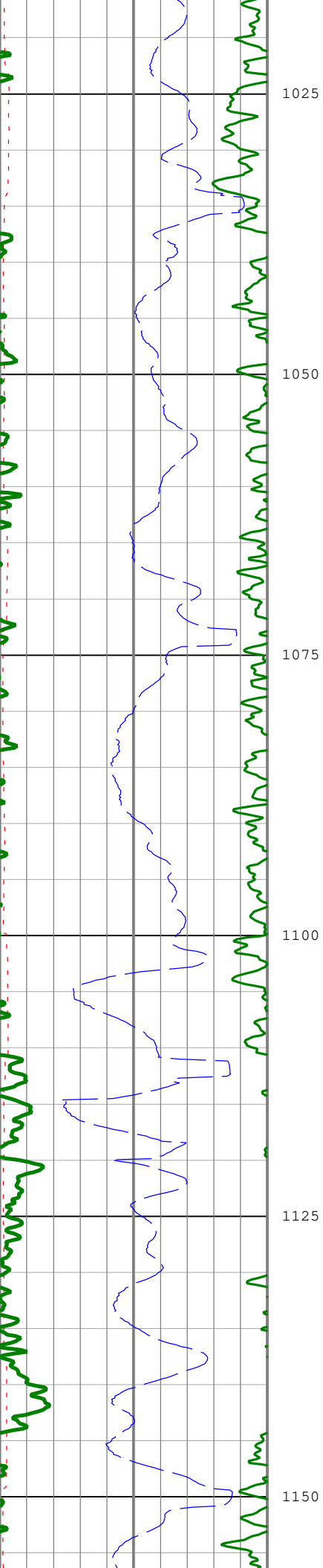
Run 7: Toolstring			Run 7: Remarks
Equip name Stabilizer[2]:26 068	Length 28.34	MP name 17 1/4 in. Non-Mag Stab	All depths referenced to driller's depth. See EOWR for depth tracking details.
			All data from tool memory. All data acquired while drilling.
			Gamma Ray measurement is corrected for collar thickness only and not for Potassium content.
NMDC: 9 1/2"[2]:26105-1	26.00	Non-Mag Pony Collar	Resistivity measurements are borehole compensated and environmentally corrected for bit size, mud resistivity and temperature.
			Run 7 and 12 1/4 x 17 1/2 in. section TD at 1906 m.
TELE900:G509 3 MSSU900:71539 Upper Extender:2 6-02-16 MDC900:G5093 MMA:2068 MIDI:3956 PMGR:455 PME:A:3708 MTA:2526 MTK900:008/298 895-2142 MSS900:973247	23.35	TeleScope	

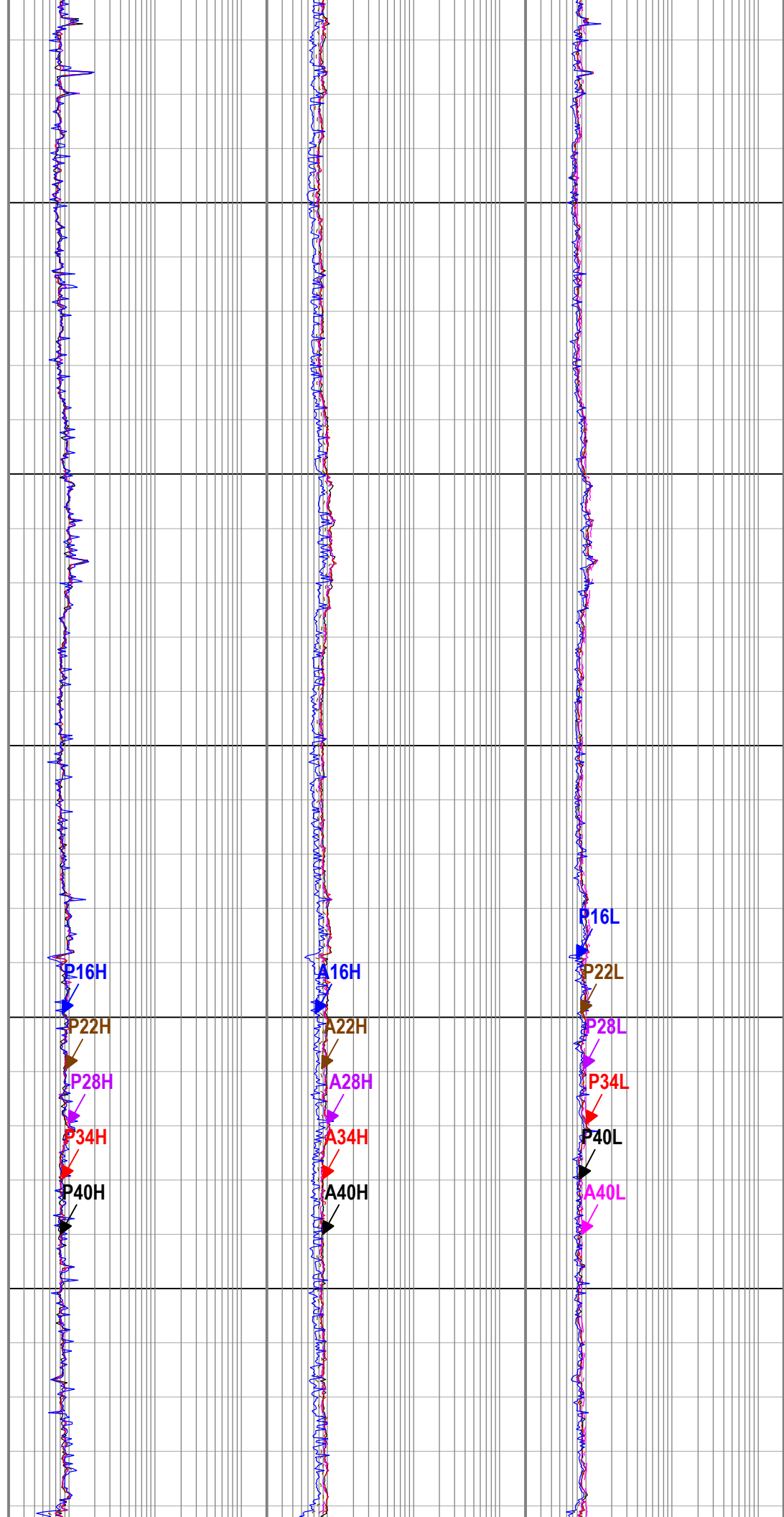
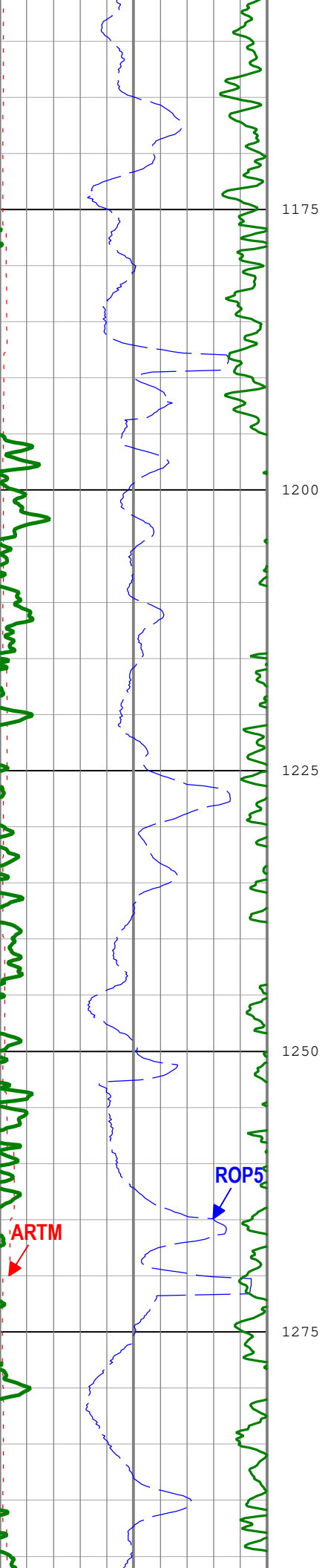


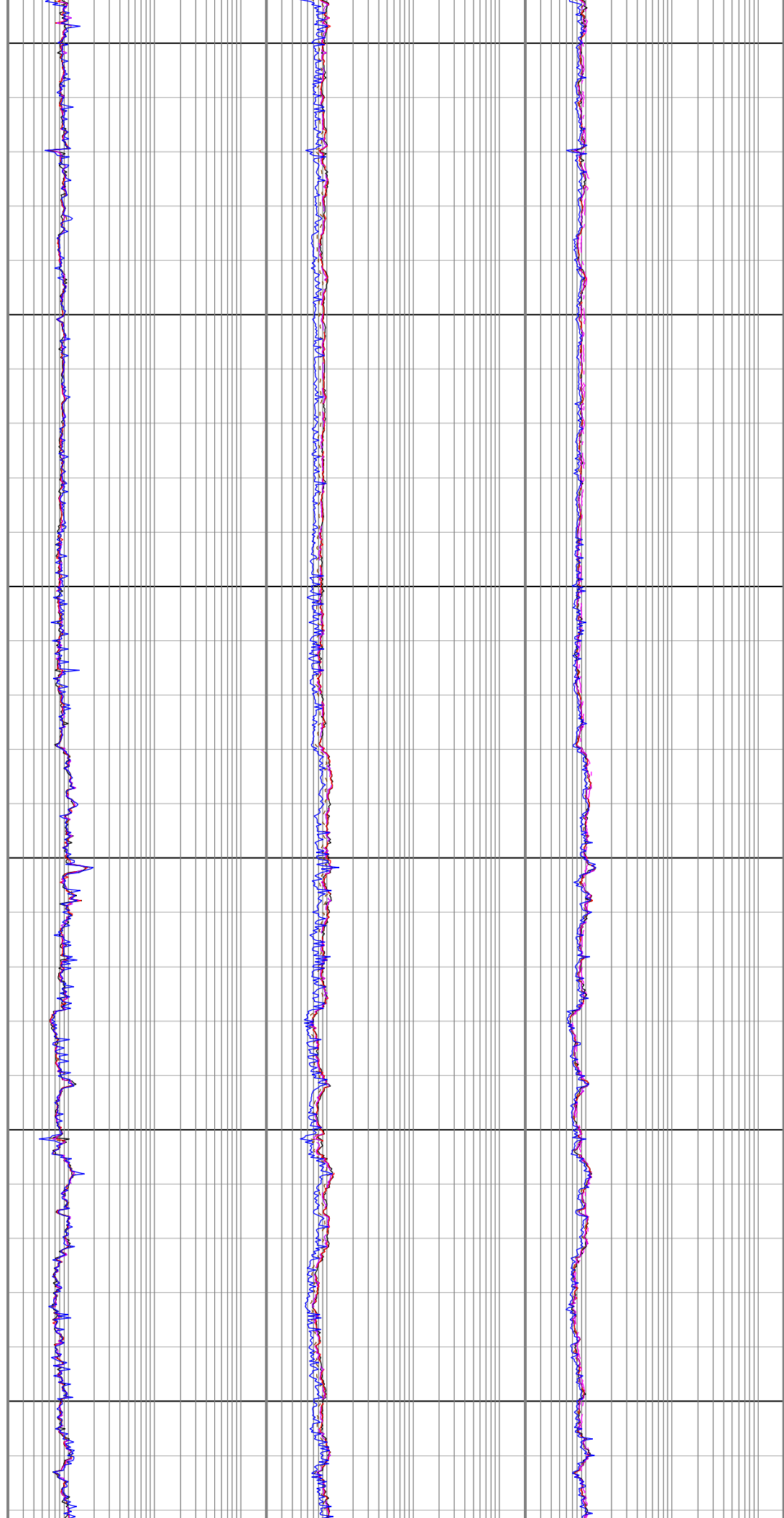
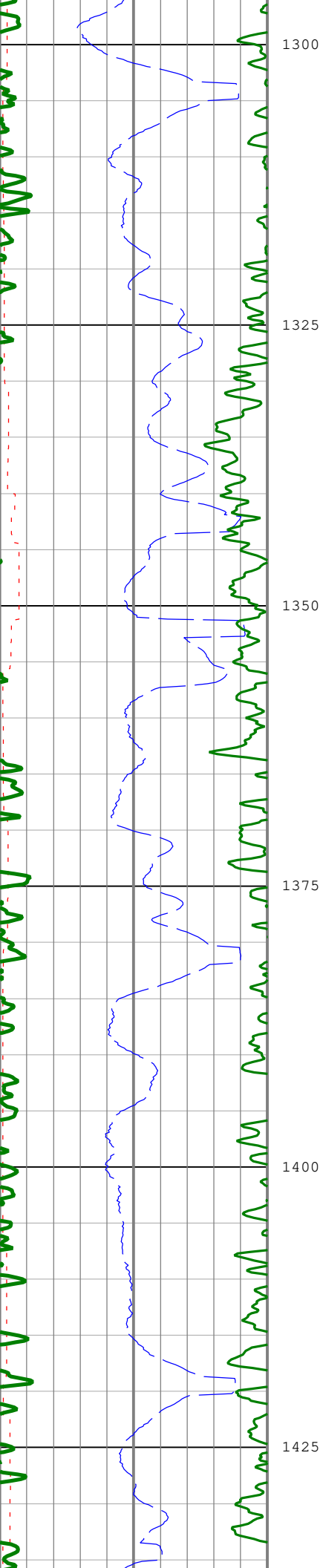
Phase Shift Resistivity 34 inch
Spacing at 400 KHz,
Environmentally Corrected. (P34L)
ARC9 RM

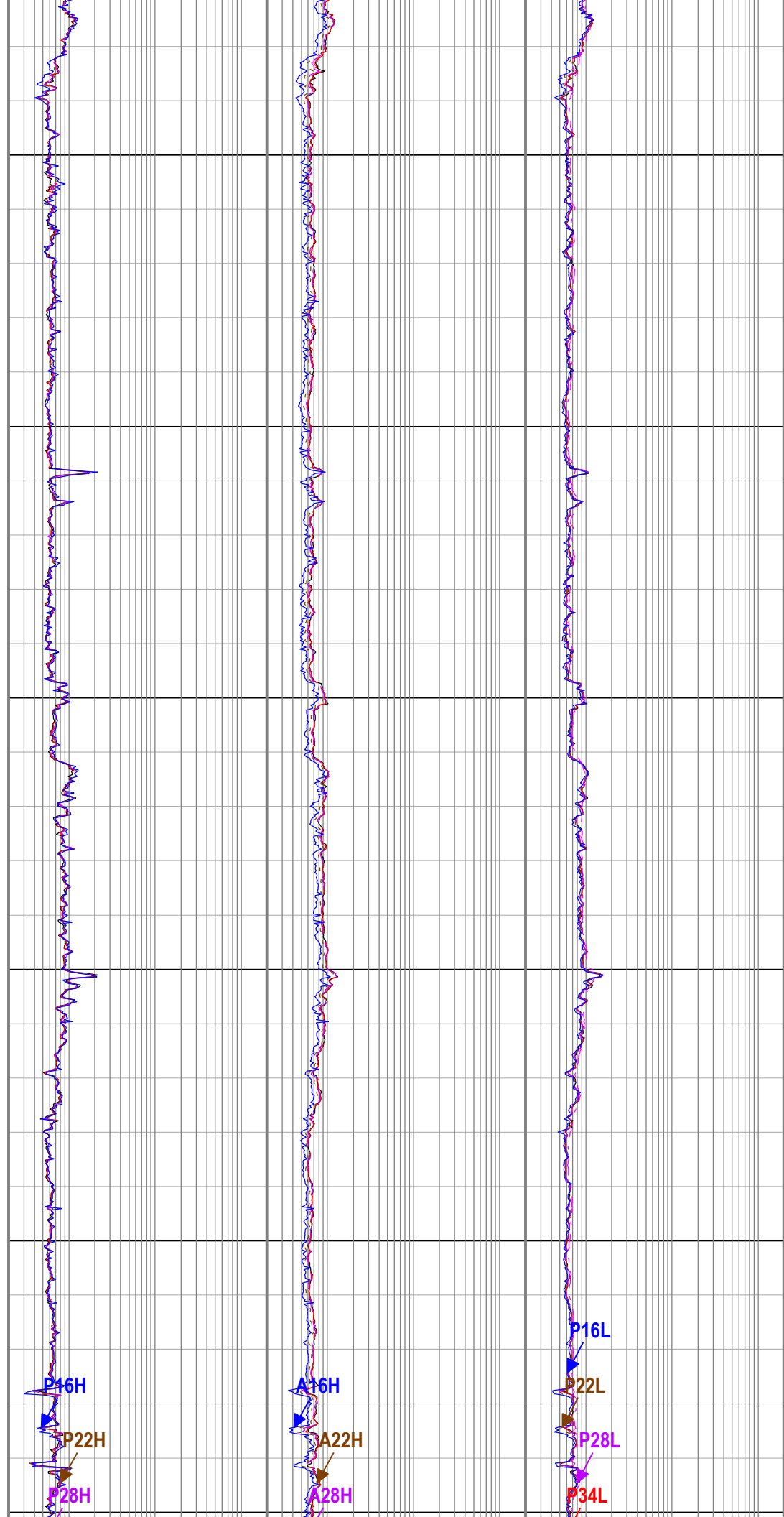
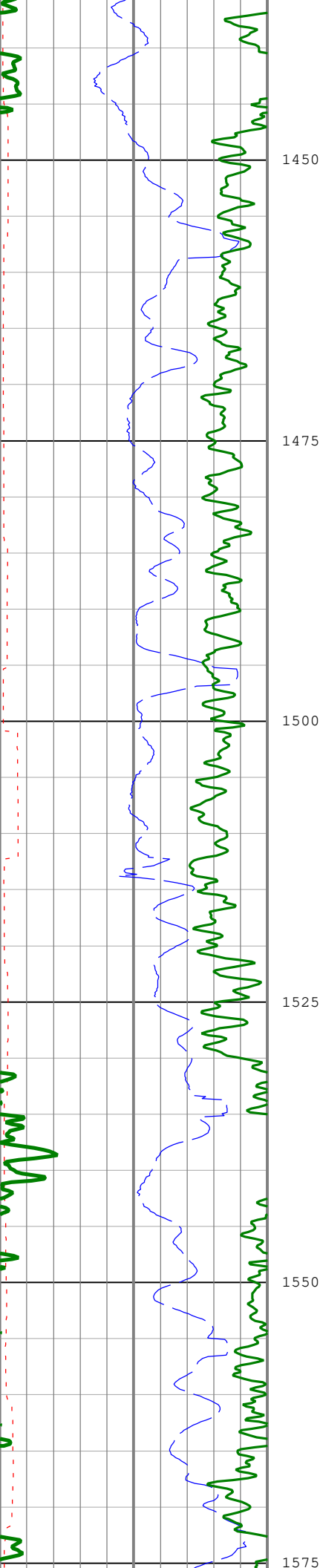
Phase Shift Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (P40H) ARC9 RM			Attenuation Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (A40H) ARC9 RM		
0.2	ohm.m	200	0.2	ohm.m	200
Phase Shift Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected. (P34H) ARC9 RM			Attenuation Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected. (A34H) ARC9 RM		
0.2	ohm.m	200	0.2	ohm.m	200

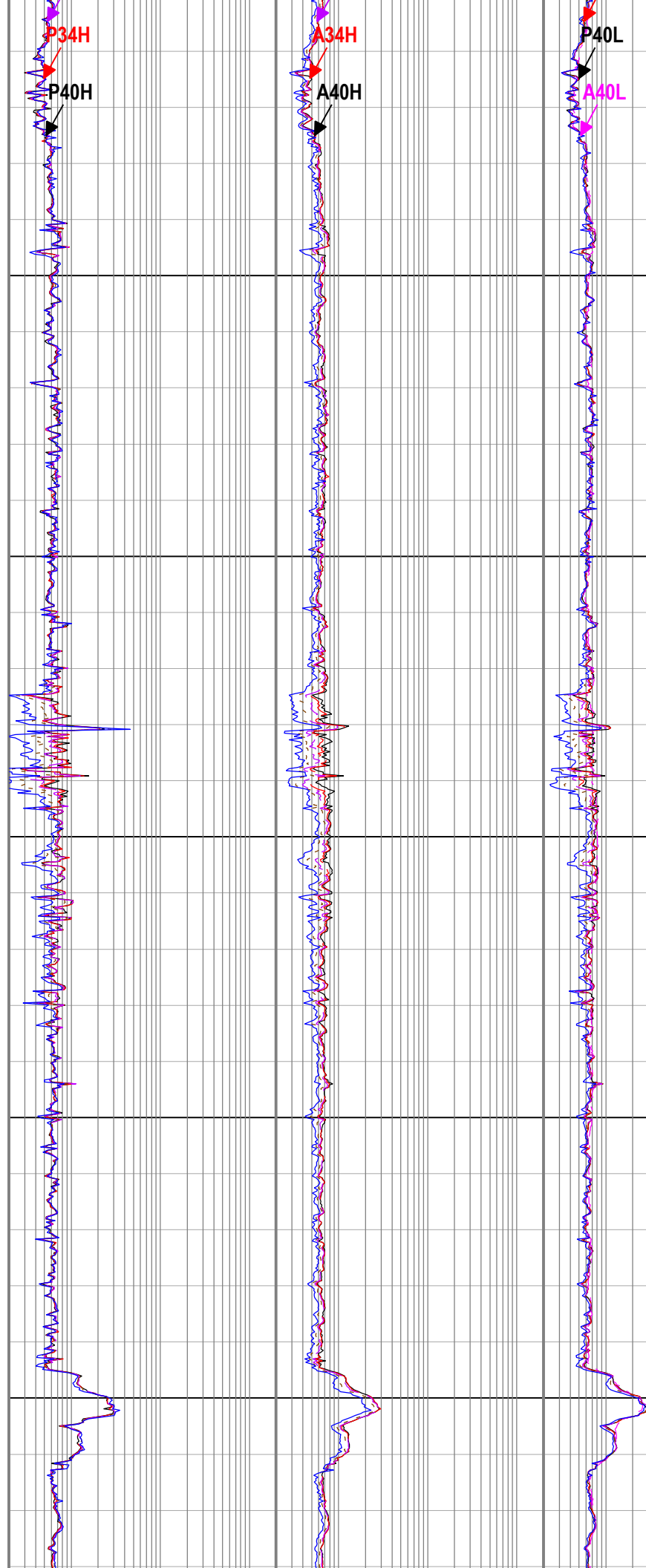
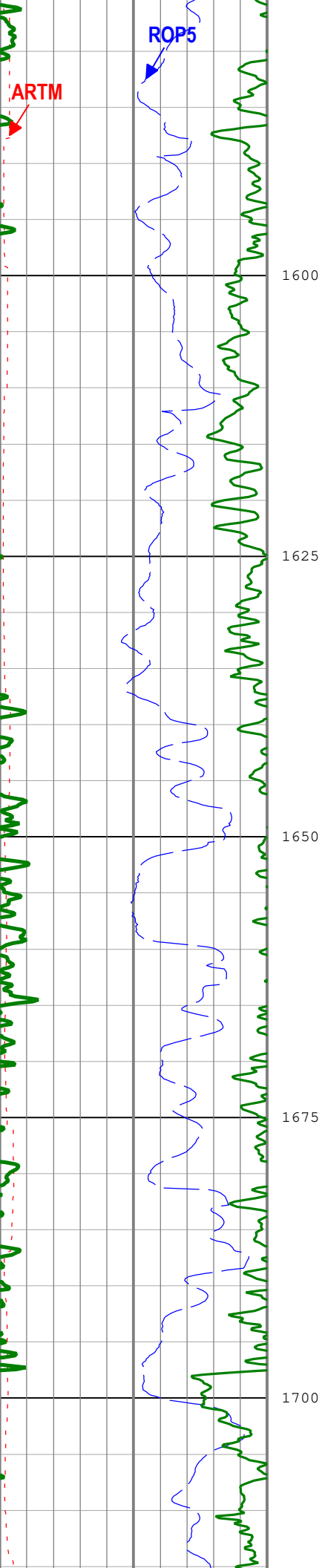


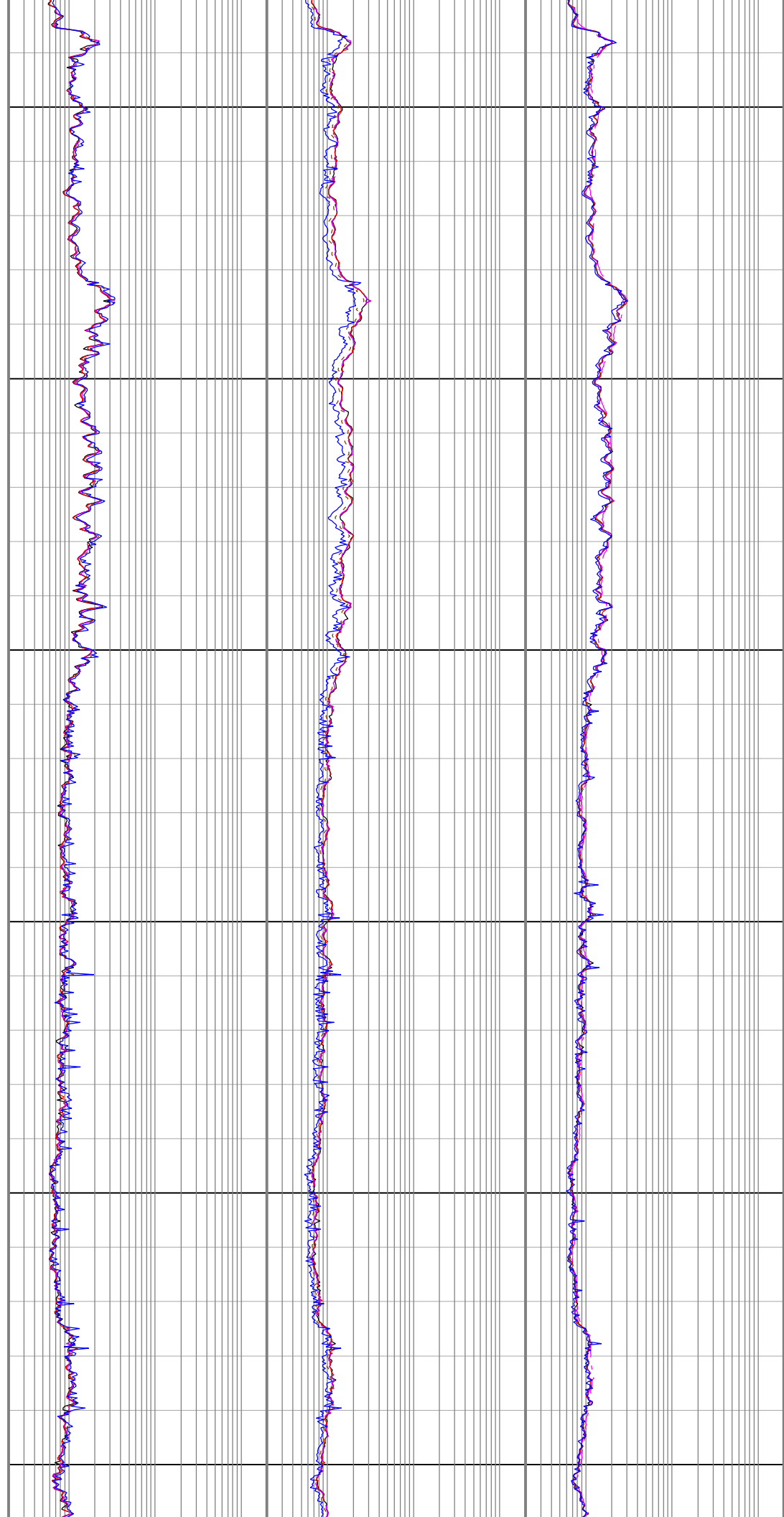
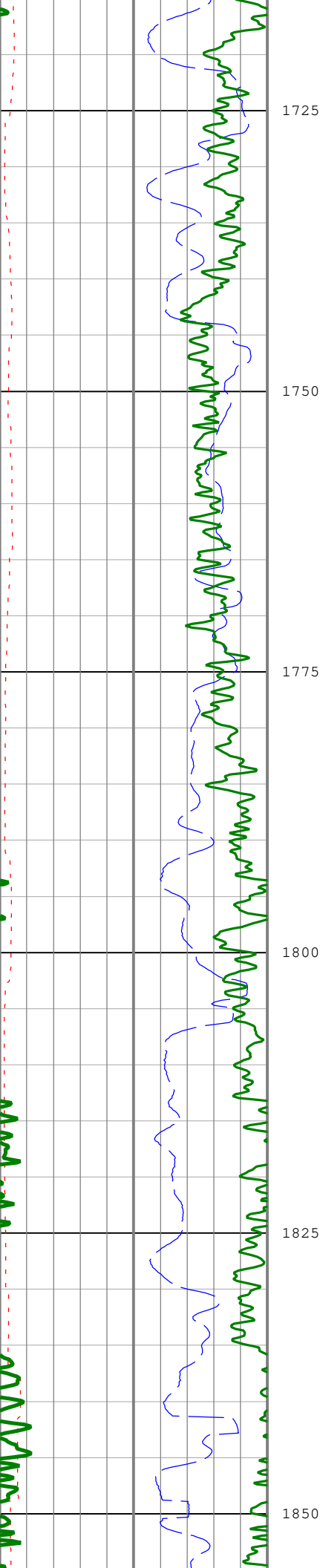


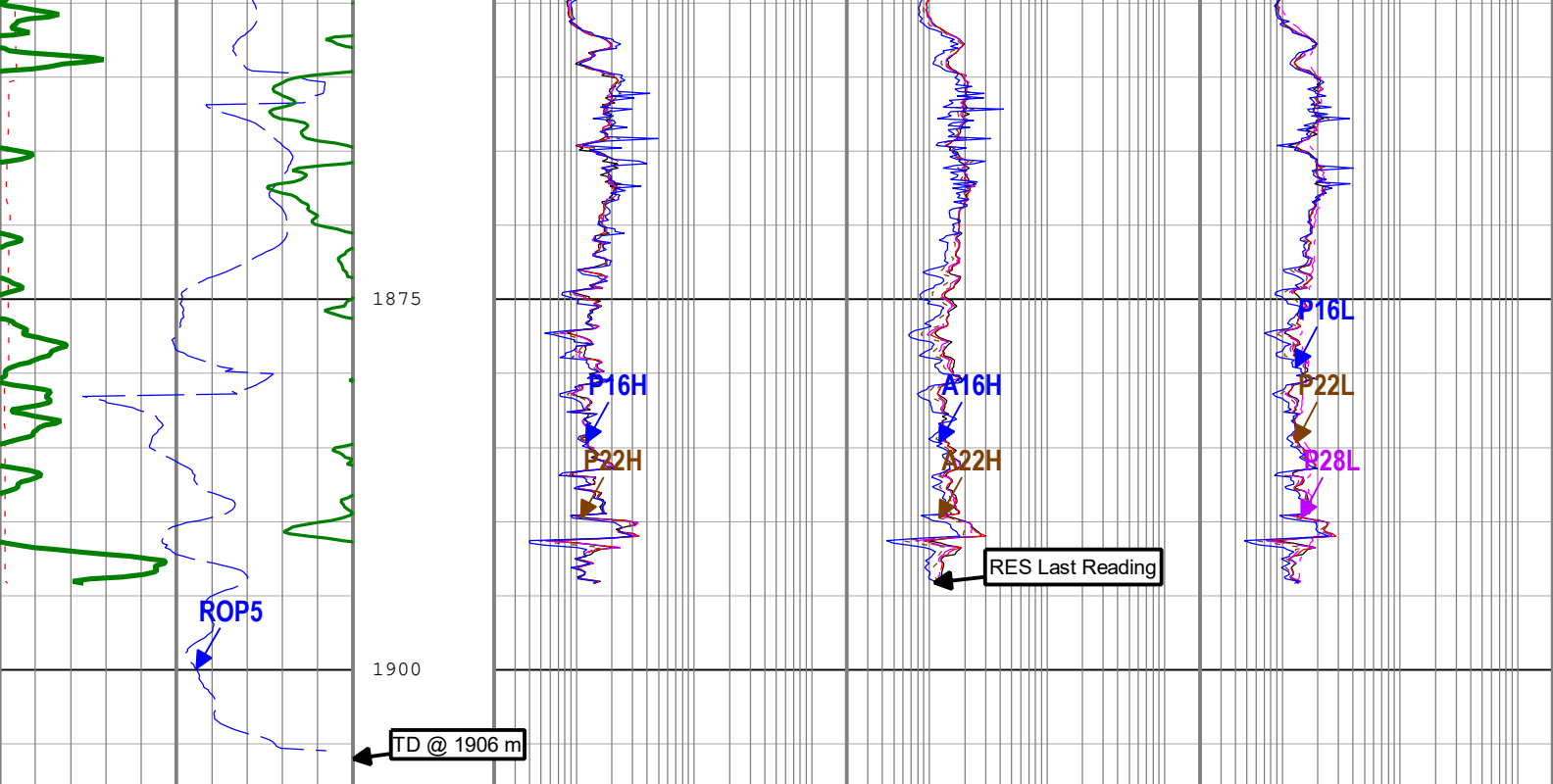












Resistivity Time After Bit (ARTM) ARC9 0-----h-----10	Phase Shift Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (P40H) ARC9 RM 0.2 ohm.m 200	Attenuation Resistivity 40 inch Spacing at 2 MHz, Environmentally Corrected. (A40H) ARC9 RM 0.2 ohm.m 200	Attenuation Resistivity 40 inch Spacing at 400 KHz, Environmentally Corrected (A40L) ARC9 RM -----ohm.m-----200
Rate of penetration averaged over the last 5 ft (1.5 m) (ROP5) RT 200 m/h 0	Phase Shift Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected. (P34H) ARC9 RM 0.2 ohm.m 200	Attenuation Resistivity 34 inch Spacing at 2 MHz, Environmentally Corrected (A34H) ARC9 RM 0.2 ohm.m 200	Phase Shift Resistivity 40 inch Spacing at 400 KHz, Environmentally Corrected. (P40L) ARC9 RM 0.2 ohm.m 200
Calibrated Gamma Ray (GR_ARC_FILT) ARC9 RM 0 gAPI 150	Phase Shift Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected. (P28H) ARC9 RM 0.2 ohm.m 200	Attenuation Resistivity 28 inch Spacing at 2 MHz, Environmentally Corrected (A28H) ARC9 RM 0.2 ohm.m 200	Phase Shift Resistivity 34 inch Spacing at 400 KHz, Environmentally Corrected. (P34L) ARC9 RM 0.2 ohm.m 200
Calibrated Gamma Ray (GR_ARC_FILT) ARC9 RM 150 gAPI 300	Phase Shift Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected. (P22H) ARC9 RM 0.2 ohm.m 200	Attenuation Resistivity 22 inch Spacing at 2 MHz, Environmentally Corrected (A22H) ARC9 RM 0.2 ohm.m 200	Phase Shift Resistivity 28 inch Spacing at 400 KHz, Environmentally Corrected. (P28L) ARC9 RM 0.2 ohm.m 200
	Phase Shift Resistivity 16 inch Spacing at 2 MHz, Environmentally Corrected. (P16H) ARC9 RM 0.2 ohm.m 200	Attenuation Resistivity 16 inch Spacing at 2 MHz, Environmentally Corrected (A16H) ARC9 RM 0.2 ohm.m 200	Phase Shift Resistivity 22 inch Spacing at 400 KHz, Environmentally Corrected. (P22L) ARC9 RM 0.2 ohm.m 200
			Phase Shift Resistivity 16 inch Spacing at 400 KHz, Environmentally Corrected. (P16L) ARC9 RM 0.2 ohm.m 200

Description: Format: Log (Resistivity - Dual Frequency) Index Scale: 1:500 Index Unit: m Index Type: Measured Depth Creation Date: 23-Dec-2019 16:32:40

Run 7: Parameters

Parameter	Description	Tool	Value	Unit
ABNT	Abnormal Transmitter Indicator	ARC9	NO_TX_FAILED	
BHK	Drilling Fluid Potassium Concentration	Borehole	0	%
BS	Bit Size	DNMSESSION	Depth Zoned	in
DFD	Drilling Fluid Density	Borehole	Time Zoned	g/cm3
DFT_CATEGORY	Drilling Fluid Type	Borehole	Water	
HIGH_BLEND	High Resistivity Threshold for Blending	ARC9	2	ohm.m
LOW_BLEND	Low Resistivity Threshold for Blending	ARC9	1	ohm.m
RMS	Resistivity of Mud Sample	Borehole	0.09	ohm.m

Depth Zone Parameters

Parameter	Value	Start (m)	Stop (m)
BS	26	919	931
BS	17.5	931	1904
BS	12.25	1904	1906

All depth are actual.

Time Zone Parameters

Parameter	Value	Start Time	Stop Time	Start Depth (m)	Stop Depth (m)
DFD	1.25	19-Dec-2019 04:52:35	19-Dec-2019 17:00:00	34.44	1264.82
DFD	1.26	19-Dec-2019 17:00:00	20-Dec-2019 22:18:15	1264.82	1905.58

All depth are at tool zero.

Tool Control Parameters

Calibration Report

ARC9 (Array Resistivity Compensated 900) Calibration - Run 7

Primary Equipment :	High Pressure without AIM Receiver	AREA	8374
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RESAIRCAL - Resistivity: Air

Master (Time Frame File): 09:51:54 27-Nov-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
Attenuation T1 at 2 MHz	dB	Master	8.500	6.500	8.353	10.500	
Attenuation T2 at 2 MHz	dB	Master	6.500	4.500	6.149	8.500	
Attenuation T3 at 2 MHz	dB	Master	4.500	2.500	5.129	6.500	
Attenuation T4 at 2 MHz	dB	Master	4.600	2.600	4.142	6.600	
Attenuation T5 at 2 MHz	dB	Master	3.600	1.600	3.746	5.600	
Phase Shift T1 at 2 MHz	deg	Master	0.100	-3.900	0.713	4.100	
Phase Shift T2 at 2 MHz	deg	Master	0.100	-3.900	-0.663	4.100	
Phase Shift T3 at 2 MHz	deg	Master	0.100	-3.900	0.666	4.100	
Phase Shift T4 at 2 MHz	deg	Master	0.100	-3.900	-0.695	4.100	
Phase Shift T5 at 2 MHz	deg	Master	0.100	-3.900	0.651	4.100	
Attenuation T1 at 400 KHz	dB	Master	8.500	6.500	8.523	10.500	
Attenuation T2 at 400 KHz	dB	Master	6.500	4.500	5.988	8.500	
Attenuation T3 at 400 KHz	dB	Master	4.500	2.500	5.294	6.500	
Attenuation T4 at 400 KHz	dB	Master	4.600	2.600	3.975	6.600	
Attenuation T5 at 400 KHz	dB	Master	3.600	1.600	3.915	5.600	
Phase Shift T1 at 400 KHz	deg	Master	0.100	-3.900	0.360	4.100	
Phase Shift T2 at 400 KHz	deg	Master	0.100	-3.900	-0.415	4.100	
Phase Shift T3 at 400 KHz	deg	Master	0.100	-3.900	0.390	4.100	
Phase Shift T4 at 400 KHz	deg	Master	0.100	-3.900	-0.422	4.100	
Phase Shift T5 at 400 KHz	deg	Master	0.100	-3.900	0.384	4.100	

GRGAIN - Gamma Ray: Blanket

Master (Time Frame File): 10:35:50 27-Nov-2019

Measurement	Unit	Phase	Nominal	Low Limit	Actual	High Limit	
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Gamma Ray Calibration Gain		Master	1.000	0.640	0.979	1.200	
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Company:	Equinor	
Well:	31/5-7	
Field:	Eos	
Rig Name:	West Hercules	
Country:	Norway	

Schlumberger

VISION Resistivity
Dual Frequency
Recorded Mode